Can Photos Prove a Turbine Blade Is Damaged?

Finding 3D damage from a CAD model and a few photos, and knowing when the photos are not enough

Afeefa Azam · Khalifa University

1. Introduction: why this matters

Jet engines and gas turbines contain hundreds of blades. Small damage such as a **dent**, a **nick** on the edge or a **crack** can grow and lead to costly failures, so blades are inspected regularly [1]. Inspection is mostly done by people looking at photos or videos, which is slow and depends on the inspector. Every blade is made from a known design: its **CAD model**. So we know exactly what a healthy blade should look like. The idea of this research is to compare photos of a real blade with its healthy CAD model and find out, in 3D, **where** the blade is damaged and **how large** the damage is.

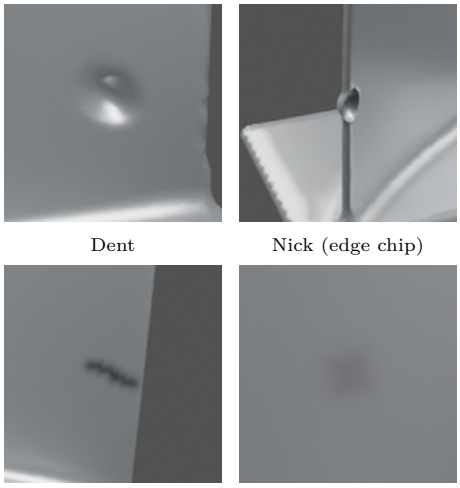


Fig. 1. Damage types we study (computer-rendered close-ups).

2. Problem statement

A dark spot in a photo can have **four different causes** (Fig. 2). Only one of them is real damage, but in a photo they can look almost the same. A 0.1 mm dent changes the image only slightly, while moving the camera by just 1° changes thousands of pixels. A computer that simply looks for "something different" will raise many **false alarms**, and a computer that always gives an answer will sometimes **invent damage** that is not there.

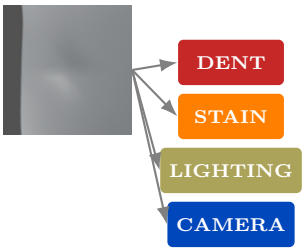


Fig. 2. One spot, four possible causes.

Research questions: (1) Using a CAD model and a few photos, can we tell real damage apart from stains, lighting changes and camera errors? (2) Can we measure the damage in 3D (location and depth in mm)? (3) Can the method recognise *when the photos are not enough* and say so, instead of guessing?

3. Objectives

Main goal: a method that reports blade damage in 3D *only when the photos can prove it*, and otherwise says "not enough evidence".

1. Separate real damage from stains, lighting changes and camera errors.
2. Locate the damage on the 3D blade and estimate its depth in millimetres.
3. Keep false alarms low and controlled (about 5% of healthy blades).
4. Recognise the type of damage: dent, nick or crack.
5. Test the method first on computer-rendered blades, then on real 3D-printed, damaged and scanned blades.

Benefit: faster, more consistent inspections that an engineer can trust, because the method states how sure it is.

4. Preliminary literature review

	2D map	3D size (mm)	Says "not sure"
Anomaly detection [2,3]	✓	×	×
CAD-based 3D copy [4]	×	×	×
3D anomaly datasets [5]	✓	×	×
This research	✓	✓	✓

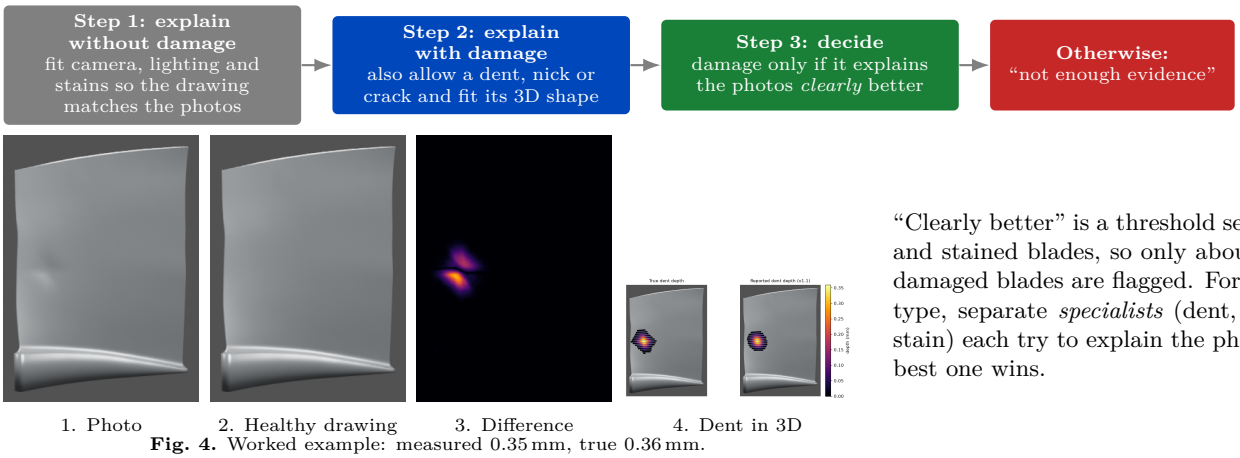
Fig. 3. What existing work provides.

Recent methods compare photos with a model of a healthy object and mark *unusual pixels*, even when the camera or lighting changes [2,3]. Others use a CAD model to build a 3D copy of an object from a few photos [4]; there, differences from the CAD are treated as noise. 3D anomaly datasets give anomaly scores, not damage sizes [5]. Our tool, "render and compare", builds on differentiable rendering [6].

The gap: no method measures damage in 3D *and* says when the photos cannot support a conclusion. An author of [3] independently suggested this question as a valuable research contribution.

5. Methodology

Main idea, "render and compare": the computer draws (renders) the healthy blade from the CAD model and compares the drawing with the photos.



"Clearly better" is a threshold set on healthy and stained blades, so only about 5% of undamaged blades are flagged. For the damage type, separate *specialists* (dent, nick, crack, stain) each try to explain the photos and the best one wins.

Fig. 4. Worked example: measured 0.35 mm, true 0.36 mm.

Test data. We built a test set of over 1,300 computer-rendered cases on public blades and six industrial blade designs (Fig. 5), with exact known damage. The test blades were kept hidden until the method was final.

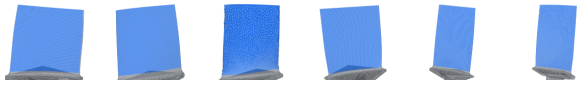


Fig. 5. The six industrial blade designs.

Preliminary results (computer-rendered blades):

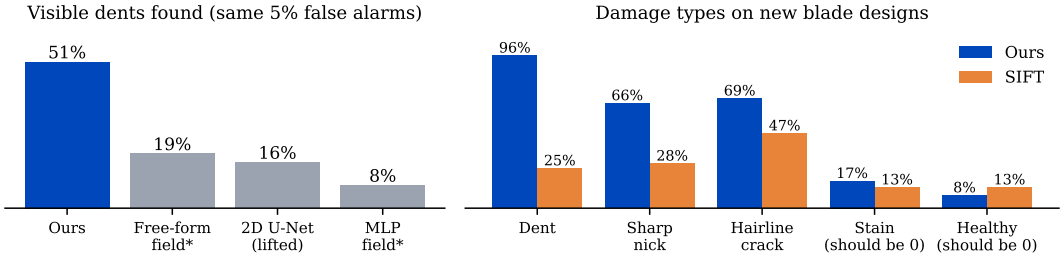
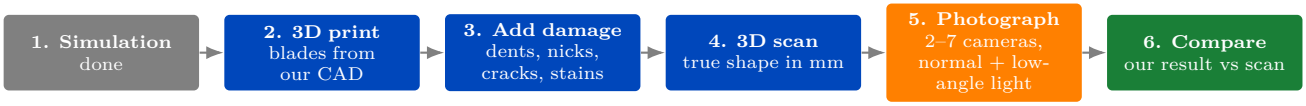


Fig. 6. Left: visible dents found at the same 5% false-alarm rate (*versions of the CAD-based method [4]). Right: damage types (dent, stain, healthy: 72 cases; sharp nick, hairline crack: 64 cases); stains and healthy blades should not be reported.

- Our method finds **51%** of visible dents at 5% false alarms; other methods find at most 19%. It finds **12 of 13** dents deeper than 0.2 mm.
- On new blade designs, false alarms dropped from 51% to **4%** after one fix. Realistic sharp nicks and hairline cracks are found in **66%** and **69%** of cases.
- A standard image-matching method (SIFT) flags only 25–47% of damage, and as many healthy blades as we do. Textbook statistical tests give 32% and 96% false alarms instead of 1%, so setting the threshold on real examples is necessary.
- **Limitation:** the location is found reliably, but on new designs depth is measured about 40% too shallow. More cameras do not fix this, so better lighting is the next step.

Next: real blades.



References

[1] BladeSynth: a rendering-based synthetic dataset for aero-engine blade defect inspection, 2025. [2] Zhou et al., PAD: a dataset and benchmark for pose-agnostic anomaly detection, 2023. [3] Yang et al., PIAD: pose and illumination agnostic anomaly detection, 2025. [4] Overdulve et al., CADSplat: sparse-view Gaussian splatting aided by CAD models, arXiv 2026. [5] Costanzino et al., SiM3D: single-instance multiview multimodal 3D anomaly detection benchmark, 2025. [6] Laine et al., Modular primitives for high-performance differentiable rendering, 2020.